Tropical cyclone - a giant heat engine

Y13 (2013) — English translation

This problem considers a simplified model of a tropical cyclone. In this model, the cyclone is considered as a giant heat engine operating according to the Carnot cycle.

Fig. 1. Diagram of air mass flows

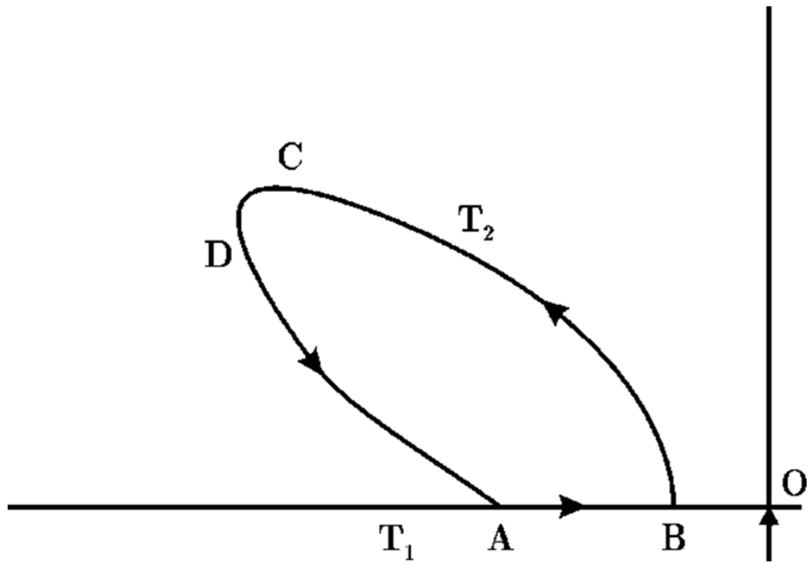

Figure 1: Figure 1.

Fig. 2. Carnot cycle

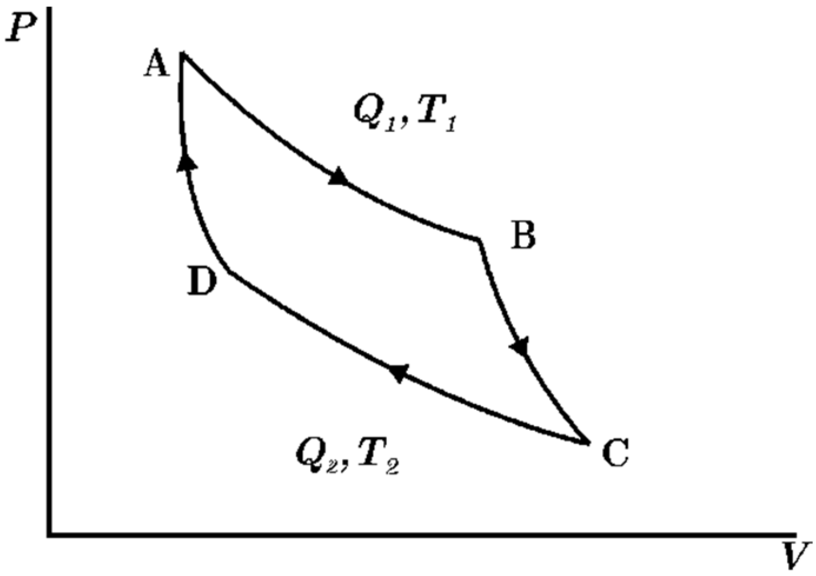

Figure 2: Figure 2.

A mass of dry air Δm flows along the ocean surface from a high pressure area (A) to a low pressure area (B) close to the center of the storm O (Fig. 1), capturing moisture (water vapor) along the way. Together with the vapors, heat Q_1 is taken from the ocean. Air masses saturated with water vapor rise into the troposphere (section BC). In the troposphere (section CD) they are discharged as a tropical shower, giving off energy Q_2.

A1	Using the first law of thermodynamics, find an expression for Q_1 (see Fig. 2) in terms of L - the specific heat of evaporation, Δq - the moisture mass of the trapped air mass Δm on the way from A to B P_A and P_B - pressure in point A and point B respectively. Use the universal gas constant R, sea surface temperature T_1, molar mass M of air.	**0**

A2	Find the total work done in one Carnot cycle by a heat engine on a mass of air Δm.	**0**

A3	Assuming that the work done by the heat engine goes entirely to increasing the kinetic energy of the air mass, obtain an expression for the speed v_B of the air in the central part of the cyclone (the “eye” of the cyclone).	**0**

A4 Assuming $v_A = 0$, find the numerical value of v_B using the following data: **0**

- $T_1 = 273$ K, $T_2 = 213$ K;
- $R = 8.31$ J/(mol $\cdot$ K);
- relative air humidity at B $\varphi = 75\%$;
- at $T = T_1$ saturated vapor pressure $P_{sat} = 42$ mbar;
- $P_A = 1000$ mbar, $P_B = 950$ mbar;
- average molar mass of air $M = 29 \cdot 10^{-3}$ kg $\cdot$ mol^{-1};
- molar mass of water $M_\mathrm{w} = 18 \cdot 10^{-3}$ kg $\cdot$ mol^{-1};
- specific heat of evaporation of water $L = 2.26 \cdot 10^6$ J $\cdot$ kg^{-1};
- density of dry air $\rho = 1.3$ kg/m^3 (1 bar = 10^5 Pa).

A5 Assuming that the azimuthal velocity v and the radial distance r from the center of the cyclone are related by $v\sqrt{r} =$ const, and that the velocity v_A at the outer edge of the cyclone is 10 m/s, calculate the effective radius of the cyclone (assume $r_B = 10$ km). **0**

Source: `https://pho.rs/p/3971`